

This Patient Group Direction (PGD) is intended only for registered healthcare professionals who have been named and authorised by their organisation to practice under it.

Patient Group Direction
for the administration of Acetazolamide for the prevention of Altitude Sickness

Summary of NICE CKS Guidelines for Altitude Sickness

Overview

- **Altitude sickness** (acute mountain sickness or AMS) occurs when people ascend to **high altitudes too quickly, typically above 2,500 metres.**
- Caused by **hypoxia** due to reduced atmospheric pressure.

Types of Altitude Illness

1. **Acute Mountain Sickness (AMS)** – common and usually mild
2. **High-Altitude Cerebral Oedema (HACE)** – life-threatening
3. **High-Altitude Pulmonary Oedema (HAPE)** – life-threatening

Symptoms of AMS

- Develops within **6–12 hours** of ascent
- Headache (required for diagnosis) **plus** one or more:
 - Nausea or vomiting
 - Dizziness or light-headedness
 - Fatigue or weakness
 - Poor sleep
 - Loss of appetite

Prevention Advice

1. Gradual Ascent

- Ascend **no more than 300–500 metres per day** above 2,500 m
- Include a **rest day every 3–4 days** or for every 1,000 m ascent

2. Acetazolamide (for prevention)

- Consider for:
 - Previous AMS
 - Rapid or unavoidable ascent
 - High-risk individuals

Dose:

- **Acetazolamide 250 mg twice daily**, starting **1–2 days before ascent**, continue for at least **2 days at peak altitude**

Acetazolamide is **not licensed in the UK** for AMS, but widely used **off-label**.

Management of Mild AMS

- **Do not ascend further** until symptoms resolve
- Rest and monitor
- Treat headache with **paracetamol or ibuprofen**
- Mild symptoms usually resolve within **24–48 hours**

Acetazolamide for treatment

- **250 mg twice daily** can help reduce symptom duration
-

Management of Severe AMS / Suspected HACE or HAPE

- **Immediate descent** is the **priority**
- Administer **supplemental oxygen** if available
- Consider **dexamethasone** (HACE):
 - **8 mg initially**, then **4 mg every 6 hours**
- Consider **nifedipine** (HAPE):
 - **20 mg slow release BD**, only under specialist advice

These are **emergency situations** — urgent medical attention and descent are critical.

When to Seek Medical Attention

- Symptoms worsen or don't improve within 24 hours
- Neurological symptoms (e.g. confusion, ataxia)
- Breathing difficulty at rest (possible HAPE)
- Severe fatigue, reduced consciousness

Other Advice

- Stay well hydrated
- Avoid alcohol
- Avoid sleeping pills or sedatives
- Educate travellers on recognising and responding to symptoms early

Patient Group Direction

for the supply/administration of

Acetazolamide

for the treatment/prevention of

Altitude Sickness

Healthcare professionals covered and training

	Healthcare professionals allowed to work under this PGD and requirements
Qualifications and Registration	● Pharmacist registered and practicing with the GPhC ● Pharmacy technician registered and practicing with the GPhC
Training and assessment	● Pharmacists and Pharmacy Technicians must have completed training relevant to this condition to use this PGD. This training will be documented and overseen by the Get Real Health team. ● All users must be familiar with the SPC for the products as well as BNF advice and any applicable national guidelines ● All users must previously have used PGD's to supply medication ● Must work in compliance with the SOPs of their own employer ● All users must have appropriate indemnity in place to cover this service
Further training and responsibilities	● All users must be up to date with CPD requirements and appraisal ● All users must be up to date with MHRA and safety alerts with regards to medical products ● All users must understand the concepts of capacity and consent and be aware of the Mental Capacity Act 2005.

PGD Indication	Acetazolamide is indicated for the prevention and treatment of acute mountain sickness (AMS) in adults travelling to altitudes typically above 2,500 metres. For prevention, treatment is started prior to ascent. For treatment, it is initiated upon symptom onset.
Inclusion criteria	- Adults aged 18 years and over. - Travelling to or currently at altitudes above 2,500 metres. - Requesting preventative treatment or symptomatic treatment for AMS. - Informed consent obtained. - No contraindications to acetazolamide therapy.
Exclusion criteria	- Known hypersensitivity to acetazolamide or sulfonamides. - Severe renal or hepatic impairment. - History of electrolyte imbalance. - Pregnancy or breastfeeding. - Concurrent use of high-dose aspirin or other interacting medications.
Cautions	- Use with caution in patients with mild renal impairment. - Monitor for symptoms of electrolyte imbalance and dehydration. - Advise patient to maintain adequate hydration and avoid alcohol. - Discontinue if severe side effects develop.
Actions if patient is excluded or declines treatment	Advise on alternative treatment options and how these can be accessed. Document any advice given and the decision reached. Inform or refer to the GP as appropriate

Details of the medicine

Name, form and strength of medicine	Acetazolamide
Legal category	POM
Storage	Store below 25°C in a dry place.
Route/method of administration	Oral administration, with or without food.
Dose and frequency	Prevention: 125 mg twice daily starting 1–2 days before ascent and continuing for 2–5 days. Treatment: 250 mg twice daily for up to 3 days.
Quantity to be administered and/or supplied	As required based on prevention or treatment protocol
Maximum or minimum treatment period	Duration depends on altitude exposure and clinical need.
Adverse effects	Paresthesia, polyuria, taste alteration, nausea, drowsiness. Rare: electrolyte imbalance, kidney stones, blood dyscrasias. Refer to SmPC for full details.

Yellow card reporting	Report suspected adverse effects via https://yellowcard.mhra.gov.uk and
-----------------------	--

	advise the traveller to consult a healthcare professional if symptoms worsen.
Records to be kept	Record the following information: ● that valid informed consent was given ● name of individual, address, date of birth and GP ● name of healthcare practitioner ● name and brand of medication ● date of supply dose, form and route quantity supplied ● advice given, including advice given if excluded or declines treatment ● details of any adverse drug reactions and actions taken supplied via PGD Records should be signed and dated. All records should be clear, legible and contemporaneous. Electronic records can be made. A record of all individuals receiving treatment under this PGD should also be kept for audit purposes as follows: ● Adults aged 18 years and over - keep records for 8 years ● Children aged under 18 years -keep records until the 25th birthday (if the patient was 17 years when treatment was finished, keep records until the 26th birthday)

Patient information

Written information to be given to patient or carer	Supply the patient information leaflet (PIL) provided with the medication.
Follow-up advice to be given to patient or carer	To seek medical advice if symptoms worsen rapidly or significantly, do not improve in 3–4 weeks, or they become systemically very unwell.

Key references

- NICE Medicines Practice Guideline 2 (MPG2): Patient Group Directions. Published March 2017 <https://www.nice.org.uk/guidance/mpg2>
- NICE MPG2 Patient group directions: competency framework for health professionals using patient group directions. Updated March 2017 <https://www.nice.org.uk/guidance/mpg2/resources>
- NICE CKS Guidance
- British National Formulary (BNF) Key information on the selection, prescribing, dispensing and administration of medicines [BNF \(British National Formulary\) | NICE](#)
- Summary of Product Characteristics

Agreement to practise by the Registered Healthcare Professional

By signing this PGD you are confirming that you agree to its contents and that you will work within it. PGDs do not remove inherent professional obligations or accountability. It is the responsibility of each healthcare professional to practise only within the bounds of their own competence and professional code of conduct.

Name and address of pharmacy premises / clinic premises to which this PGD relates

Name of healthcare professional	Job title	Registration number	Signature
Valid from date		Expiry date	

PGD adoption and authorisation by the employer/clinical lead

The employer has ensured that the person using this PGD, has the knowledge and skills to safely provide the service which will encompass the supply or administration of a medication or vaccine.

This section must be signed by the superintendent or clinical lead responsible for this service.

Name of superintendent / clinical lead	Job title & name of organisation	Signature	Date

Name of professional group signatory

Job title & name of organisation

Signature

Date

Signed on behalf of Get Real Health

Name	Qualification	Signature	Date
Nitin Shori	Medical Doctor GMC: 6047293		25/8/2025
Chris Pilkington	Pharmacist GPhC: 2046322		25/8/2025

Change history

Version number	Change details	Date
001	Development & issue of new PGD	25 th June 2025